

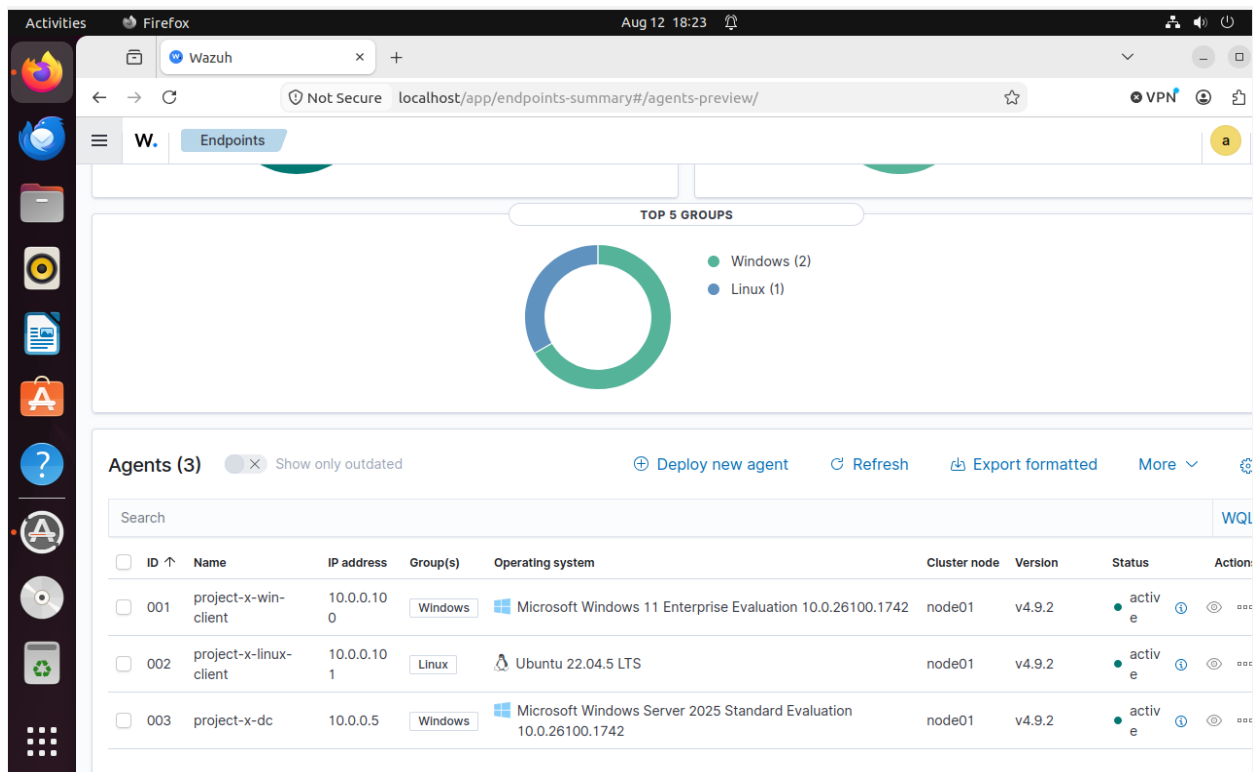
Wazuh Detection Configuration

Wazuh is an open-source platform that provides extended detection response (XDR) and System Information and Event Management (SIEM). This project will showcase the creation of essential alerts detections. I will then demonstrate our alert detections through my network environment to simulate a possible threat that may need to be investigated.

SIEMS and XDR provide the following: Event Correlation, Real-time Alerts Intrusion Detection, and Endpoint Visibility

First, I created agent groups based on operating systems. These workstations will be connected to Wazuh to analyze data received from the agents. Wazuh will be my primary hub for security defense that will be used to log data and analyze traffic on the network.

The image below shows the agents connected to the group workstations (Linux, Windows)



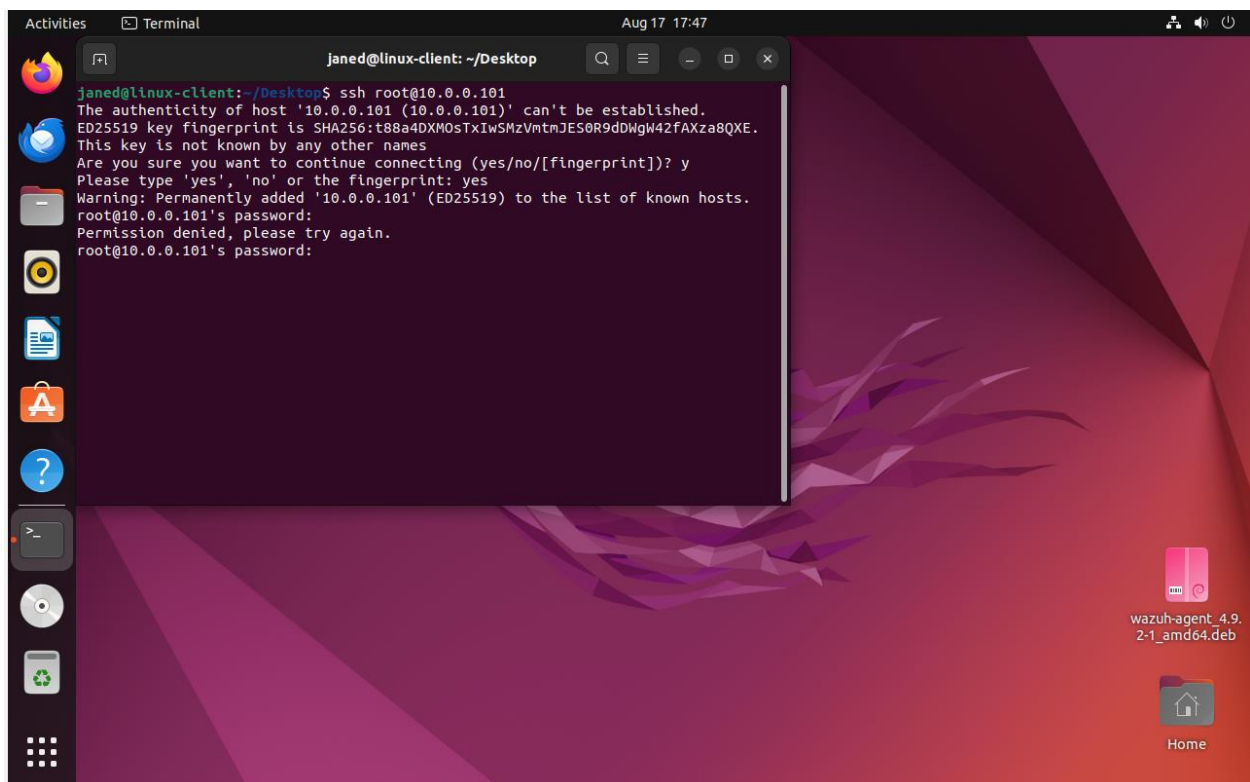
Authentication Attempts successful & Failures Through SSHD

Default Detection Alert Example

Wazuh has a default built-in rule detection for authentication failures. Below is an example of conducting a failed login attempt to demonstrate the alert on Wazuh. Notice the Timestamp Aug 17th @ 17:47

The first screenshot is the failure authentication attempt

The second screenshot is the alert of Wazuh detecting the activity (note the timestamp and log notification for failed password).



The screenshot shows the Wazuh Discover interface in a Firefox browser. The left sidebar contains a search bar with 'wazuh-alerts-*' and a list of available fields. The main panel displays a table of alert data for a failed SSH login.

Available fields:

- _source
- _index
- agent.id
- agent.ip
- agent.name
- data.arch
- data.command
- data.dpkg_status
- data.dstuser
- data.extra_data
- data.file
- data.gid
- data.home
- data.package
- data.pwd

Alert Data:

Field	Value
_index	wazuh-alerts-4.x-2026.08.17
agent.id	002
agent.ip	10.0.0.101
agent.name	project-x-linux-client
data.dstuser	root
data.srcip	10.0.0.101
data.srport	46492
decoder.name	sshd
decoder.parent	sshd
full_log	Aug 17 21:47:38 linux-client sshd[13891]: Failed password for root from 10.0.0.101 port 46492 ssh2
id	1787003259.1876392
input.type	log
location	journald
manager.name	sec-box
predecoder.hostname	linux-client
predecoder.program_name	sshd
predecoder.timestamp	Aug 17 21:47:38
rule.description	sshd: authentication failed.

Detection Alert Creations

This section will be for the creation of my own alerts. My first alert will be triggered when 3 authentication attempts fail. Below is the screenshot of the creation. The screenshot below shows the creation.

The screenshot shows the Wazuh alerting interface in a Firefox browser window. The address bar shows the URL: `localhost/app/alerting#/monitors/mxPDEaABaz50EuMCbH0D?alertState=ALL&from`. The interface has a sidebar with various icons and a main content area. The main content area is titled "3 Failed SSH Attempts" and includes a status indicator "Enabled" and buttons for "Edit", "Disable", "Export as JSON", and "Delete".

Overview

Monitor type	Monitor definition type	Data sources	Total active alerts
Per query monitor	Visual Graph	wazuh-alerts-4.x-*	0

Schedule	Last updated	Monitor ID	Monitor version number
Every 1 minutes	08/17/26 6:06 pm EDT	mxPDEaABaz50EuMCbH0D	1

Associations with composite monitors

Triggers (1)

Name ↑	Number of actions	Severity
3 Failed SSH Attempts	0	3

History

08/15/2026 12:00 AM → 08/17/2026 06:06 PM

Another alert will be for WinRM to detect successful and unsuccessful authentication attempts into a Windows machine. The next alert will be modifications of files. The screenshot below shows the alerts created. The next lab will demonstrate these alerts along with the basic wazuh alerts configurations.

The screenshot shows the Wazuh Alerts Monitors page in a Firefox browser. The browser's address bar shows the URL: `localhost/app/alerting#/monitors?from=0&search=&size=20&sortDirection=desc&`. The page has a navigation bar with 'Alerting' and 'Monitors' tabs. Below the navigation bar, there are tabs for 'Alerts', 'Monitors', and 'Destinations'. The 'Monitors' tab is active, displaying a table of monitors. The table has columns: Monitor n..., State, Type, Latest alert, Last notificati..., Active, Acknowledged, Errors, Ignored, Associations ..., and Actions. There are three monitors listed: 'WinRM Logon', 'File Accessed', and '3 Failed SSH Attempts'. Each monitor is 'Enabled' and has a 'Per query' type. The 'WinRM Logon' monitor has a latest alert of 'WinRM logon' and a last notification of '08/18/26 7:24 pm'. The 'File Accessed' and '3 Failed SSH Attempts' monitors have '--' for the latest alert and '-' for the last notification. The 'Active' column shows 1 for 'WinRM Logon' and 0 for the other two. The 'Acknowledged', 'Errors', and 'Ignored' columns all show 0 for all monitors. The 'Associations ...' column shows 0 for all monitors. The 'Actions' column shows '...' for all monitors. At the bottom of the table, there is a 'Rows per page: 20' dropdown and a pagination control showing '< 1 >'. The browser's status bar at the bottom shows 'Aug 18 19:25'.

Monitor n...	State	Type	Latest alert	Last notificati...	Active	Acknowledged	Errors	Ignored	Associations ...	Actions
☐ WinRM Logon	Enabled	Per query	WinRM logon	08/18/26 7:24 pm	1	0	0	0	0	...
☐ File Accessed	Enabled	Per query	--	-	0	0	0	0	0	...
☐ 3 Failed SSH Attempts	Enabled	Per query	--	-	0	0	0	0	0	...